Topic 4: API Authentication and Security

N8N Hands-On API Integration Course • Step-by-Step Practical Guide

> **Overview:** This practical guide walks you through creating reusable credentials, configuring Basic Auth (base64 username/password), setting up API Key/Header Auth, connecting to OAuth 2.0 endpoints using client credentials and redirect callbacks, and analyzing JSON Web Token (JWT) structures.

Step 1: Manage Reusable Credentials in the n8n Console

n8n stores credentials securely separate from workflows to prevent plain-text exposure:

1. Open your n8n Dashboard.
2. In the left navigation menu, click on **Credentials**.
3. Click the **Add Credential** button in the top-right corner.
4. Search for **Basic Auth** and click **Create**.
5. Configure:
   - **User Name:** `postman`
   - **Password:** `password`
6. Click **Save** to encrypt and store your credentials.

Step 2: Implement Basic Authentication in HTTP Request Node

Test Basic Auth credentials against the secure Postman Echo mock endpoint:

1. Add a **Manual Trigger** node to your canvas.
2. Connect a new **HTTP Request** node named `HTTP_BasicAuth`.
3. Set parameters:
   - **Method:** `GET`
   - **URL:** `https://postman-echo.com/basic-auth`
   - **Authentication:** `Generic Credential Type` → `Basic Auth`
4. Click **Test step**. Verify status `200 OK` and `"authenticated": true`.

Step 3: Implement Header-Based Token Authentication (Bearer Token)

Pass Bearer tokens inside the HTTP Authorization header:

1. Add a new **HTTP Request** node named `HTTP_BearerTokenAuth`.
2. Set parameters:
   - **Method:** `GET`
   - **URL:** `https://postman-echo.com/headers`
   - **Authentication:** `Generic Credential Type` → `Header Auth`
   - **Name:** `Authorization` | **Value:** `Bearer my_secret_token_9988`
3. Click **Test step**. Confirm the echoed response contains: `"authorization": "Bearer my_secret_token_9988"`.

Step 4: Configure OAuth 2.0 Consent Handshake

OAuth 2.0 uses a secure redirection handshake to authorize access without sharing passwords:

1. Add a new HTTP Request node and select **Predefined Credential Type** → OAuth 2.0 provider.
2. Copy n8n's callback Redirect URL: `https://yourdomain.n8n.cloud/rest/oauth2-credential/callback`.
3. Register the app in your provider's developer console (e.g. GitHub Developer Settings) and paste the callback URL.
4. Copy the generated **Client ID** and **Client Secret** into n8n.
5. Click **Connect** / **Authorize** to activate automated token management and background refreshes.

Step 5: Deconstruct JSON Web Tokens (JWT)

A JSON Web Token consists of three parts separated by dots (`Header.Payload.Signature`):

1. Inspect the mock JWT string:

```
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiIxMjM0NTY3ODkwIiwibmFtZSI6IkpvaG4gRG9lIiwiYWRtaW4iOnRydWV9.TJVA95OrM7E2cBab30RMzE5
```

2. Navigate to `https://jwt.io/` and paste the string to analyze:
   - **Header (Red):** Metadata including algorithm type (`HS256`) and token type (`JWT`).
   - **Payload (Blue):** Claims data: `sub` (`1234567890`), `name` (`John Doe`), and `admin` (`true`).
   - **Signature (Green):** Cryptographic verification hash.

Step 6: Save and Execute the Full Workflow

1. Rename the workflow to: **N8N API Authentication Workflow**.
2. Click **Save** and then **Test workflow**. Confirm all nodes execute displaying green checkmarks.